Doan Van Tai

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Profile

Final-year Computer Networks and Communications student at UIT, VNU-HCM, working toward a career as a Research Engineer. Experience building network testbeds, adaptive routing systems, and cloud infrastructure; currently exploring AI for Security at UIT InSecLab.

Education

University of Information Technology, VNU-HCM 2023 to 2027 (expected)

B.Eng. in Computer Networks and Communications · GPA 3.52 / 4.00

Research & Lab Experience

Research Intern, AI for Security Nov 2026 to Present

UIT InSecLab · University of Information Technology, VNU-HCM

Exploring large language models and AI agents for security-oriented applications, including how agents select and use tools or skills for different tasks. Current learning topics include retrieval and reinforcement learning. This work is exploratory.

Network Digital Twin Research, Project Lead 2025 to 2026

UIT-supported undergraduate research · Advised by Dr. Huynh Van Dang

- Built a Mininet / Ryu testbed synchronized bidirectionally with Eclipse Ditto; archived experiments measured upward sync p95 at 1.017 s over 30 trials.
- Implemented anomaly detection, what-if analysis, and closed-loop mitigation. The stored controller A/B study measured h3 throughput at 0.295 Mbps OFF versus 2.146 Mbps ON under UDP flood (32 trials, 8 blocks).
- Preserved experiment artifacts and offline reproduction scripts. Detector limitations and the what-if operating region are documented in the project.

State-Model-Regret for Network Routing 2026 to Present

Research Advisor: Dr. Huynh Van Dang

Ongoing study of how current-state information and action-response modeling affect expected routing decision regret. Uses controlled queueing models and simulation; findings remain exploratory.

Selected Engineering Projects

Deep Reinforcement Learning for SDN Routing

PyTorch · Gymnasium · Mininet · Ryu · Vue / Flask

- Implemented Dueling / Double DQN with prioritized replay, an 81-dimensional state, action masking, and curriculum learning.
- Connected the agent to an SDN testbed and an OSPF comparison pipeline across congestion and failure scenarios.

Cloud Infrastructure & GitOps on AWS (Team Project)

Terraform · EKS · GitHub Actions · ArgoCD · Prometheus

- Contributed infrastructure as code and GitOps delivery for a three-tier application, with Blue/Green rollout analysis, observability, and security controls.

Technical Skills

Networks & digital twins: Mininet · Ryu · OpenFlow · Eclipse Ditto · TCP/IP · Linux

Learning & experimentation: Python · PyTorch · Gymnasium · scikit-learn · NumPy · pandas

Cloud & engineering: AWS · Docker · Kubernetes · Terraform · ArgoCD · GitHub Actions

Certification

AWS Certified Solutions Architect (Associate) (SAA-C03), 2025.